



Species identification from photographs

Course: MLOps

Difficulty: 08/10

Detailed description

The goal of this project is to develop a **species identification service using a photograph and** to maintain it through an automated end-to-end pipeline.

iNaturalist is a citizen science platform where contributors share their observations of living organisms, documented by a photograph, location, and date; the community then collectively examines these images to identify the observation at the species level. The images are published in real time, and approximately 170 million observations, out of a total of 270 million, have achieved "research grade" status following this quality validation process.

The final application will need to allow users to submit a photograph and obtain an identification with a level of confidence, with referral to a human review when the certainty is insufficient.

The project focuses on the **production chain, not the raw performance of the model**. A simple, versioned, monitored, deployed and automatically retrainable model is infinitely better than a sophisticated model left in a notebook.

The modeling stage should remain short. A pre-trained feature extractor that is frozen, followed by a simple classification head, gives perfectly acceptable results in a few tens of minutes on a CPU. Some recommended pre-trained models: ResNet, EfficientNet, MobileNet, DINOv2, BioCLIP. The library `timm` provides access to all of these extractors with a common interface.

Note: Only photographs under an open license are reusable. Photographs in the open domain are accessible in bulk via the AWS dataset; those hosted on the static domain do not have an open license. Filtering by license and preserving attribution is part of the pipeline.

Areas to explore, by difficulty

Level 1 - Restricted perimeter, full chain

- Classification based on about ten well-represented taxa, with a fixed pre-trained extractor, and a chain from ingestion to deployment effectively closed.
- License-based filtering and attribution retention are integrated into the pipeline.

Level 2 - Scalability and cost of service

- Extension to several dozen or hundreds of taxa.
- A comparison of at least two extractors using the same protocol, systematically relating the performance gain to the service cost. It is not uncommon for the highest-performing model to be ten times more expensive for a marginal gain.
- Service optimization: latency, throughput, memory footprint, size of artifacts and container images.
- Versioning a large dataset, with a sustainable storage strategy.

Level 3 - Feedback Loop and Observability

- Closing the loop: automatic collection of validated identifications afterwards, measurement of performance in production, historical data.
- Label revision processing: what to do with past predictions when the ground truth changes?

Level 4 - Robustness and cold start

- Detection of images outside the perimeter: photographs without organisms, unknown taxa, unusable images.
- Introducing a new taxon: how many images are needed, and does the pipeline allow this to be done without manual intervention?
- Approaches with zero or very few examples for rare species.
- Active selection of observations to be examined as a priority, to maximize the gain per unit of human effort.
- Providing a visual explanation to the user, with associated latency constraints.

Data

- iNaturalist API, real-time observation feed, keyless: <https://api.inaturalist.org/v2/docs/> The limit is 60 requests per minute, with a download cap of 5 GB per hour and 24 GB per day.
- iNaturalist Licensed Observation Images, an open-source game hosted on AWS: <https://registry.opendata.aws/inaturalist-open-data/>
- Documentation and associated metadata schema: <https://github.com/inaturalist/inaturalist-open-data>
- iNaturalist developer page, including the DarwinCore archive shared with GBIF, containing validated observations published under an open license, updated weekly: <https://www.inaturalist.org/pages/developers>
- GBIF API, occurrences and media, keyless: <https://techdocs.gbif.org/en/openapi/>
- iNat 2021 Challenge - 3.3 million licensed photographs, covering 10,000 species, useful for getting started right away: https://github.com/visipedia/inat_comp/tree/master/2021
- CUB-200 or Oxford Flowers, small static games, practical for unit testing and CI: https://www.vision.caltech.edu/datasets/cub_200_2011/

Bibliography

- [\[1707.06642\] The iNaturalist Species Classification and Detection Dataset](#)- Van Horn et al., *The iNaturalist Species Classification and Detection Dataset*, reference on the long tail
- [\[1901.05555\] Class-Balanced Loss Based on Effective Number of Samples](#)- Cui et al., weighting by actual number of samples
- [\[2107.07511\] A Gentle Introduction to Conformal Prediction and Distribution-Free Uncertainty Quantification](#)- Angelopoulos & Bates, conformal prediction, to guarantee a controlled error rate
- <https://dvc.org/doc/use-cases/versioning-data-and-models>- versioning of large datasets
- [\[2009.00236\] A Survey of Deep Active Learning](#)- overview of active learning methods
- <https://huggingface.co/imageomics/bioclip-2>- a relevant model for this project